

GPU Batch Lunar Gravity Validation

Scenarios : 1
Seed : 42
Altitude range : 200-400 km
Duration : 0.0002 days
RK4 dt : 10 s
Output dt : 10 s
Dtype : float64
Truth : SH200 DOP853
GPU models : st_lrps
Frame mode : match_dynamics_engine

SH200 DOP853 is the numerical reference, not physical truth.

GPU SH200 RK4 quantifies fixed-step/framework error against that reference.

Executive Summary

Most accurate GPU model : GPU_ST_LRPS_RK4
Fastest GPU model : GPU_ST_LRPS_RK4
ST-LRPS closest by median RMS : n/a
ST-LRPS median-equivalent status: insufficient_data
ST-LRPS closest by P95 RMS : n/a

Best ST-LRPS scenario : 0
Representative ST-LRPS scenario: 0
Worst ST-LRPS scenario : 0

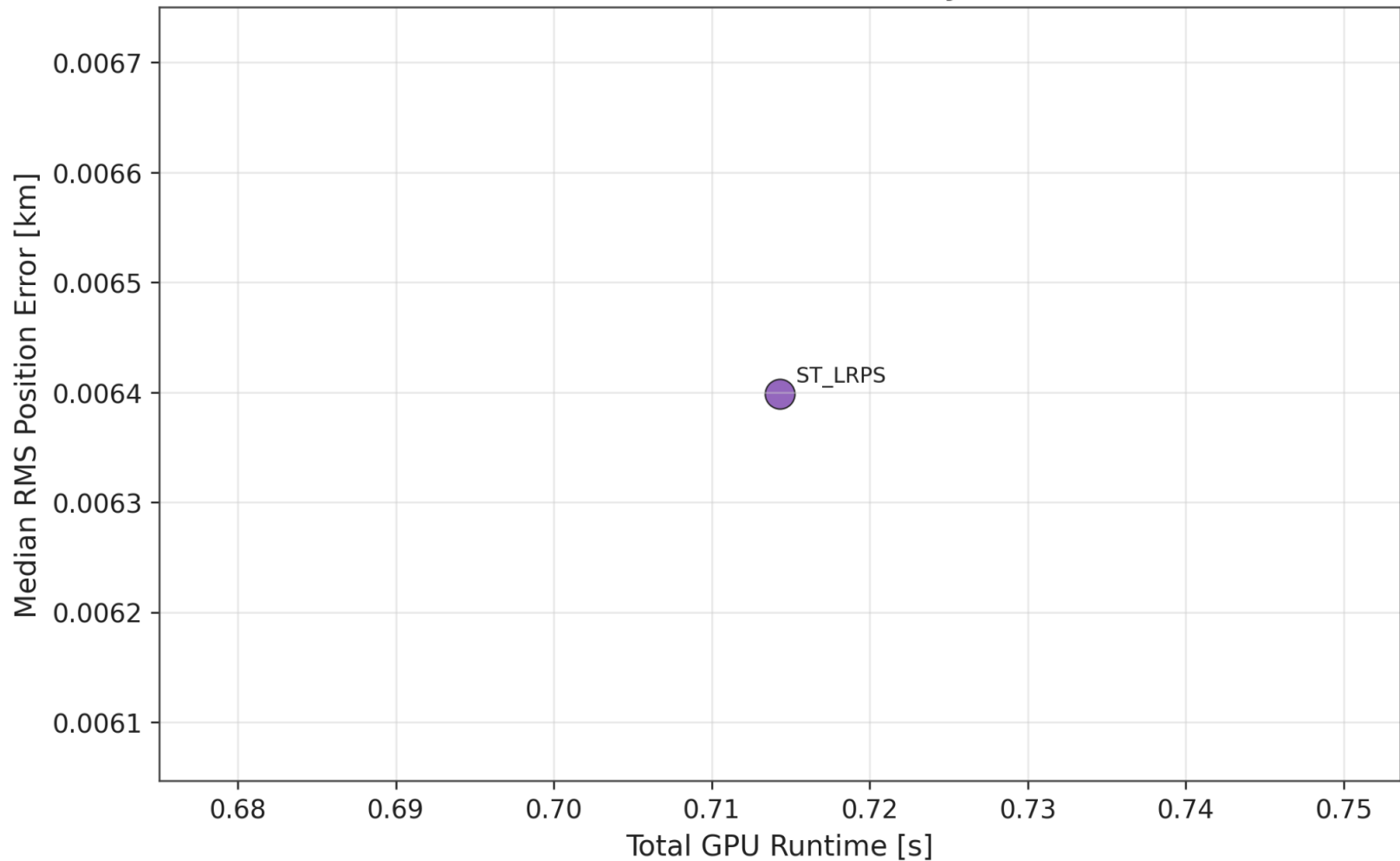
Accuracy Ranking Table

Model	Median RMS	P95 RMS	Max RMS	Median Along
GPU_ST_LRPS_RK4	0.0064	0.0064	0.0064	0.0000

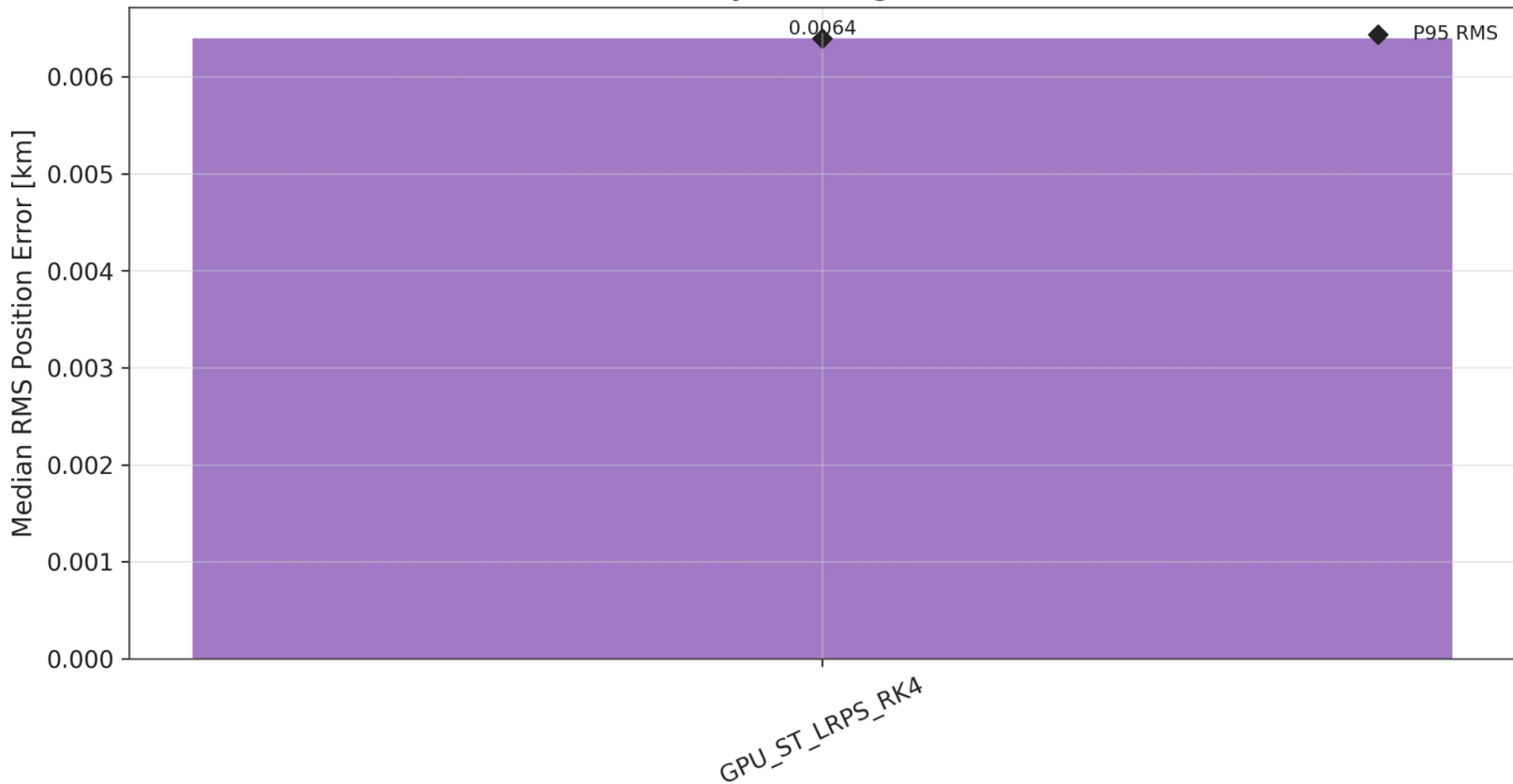
Runtime Table

Model	Runtime s	Runtime/sc	Traj-steps/s	Speedup	truth
GPU_ST_LRPS_RK4	0.714	0.71429	2.8		1.56

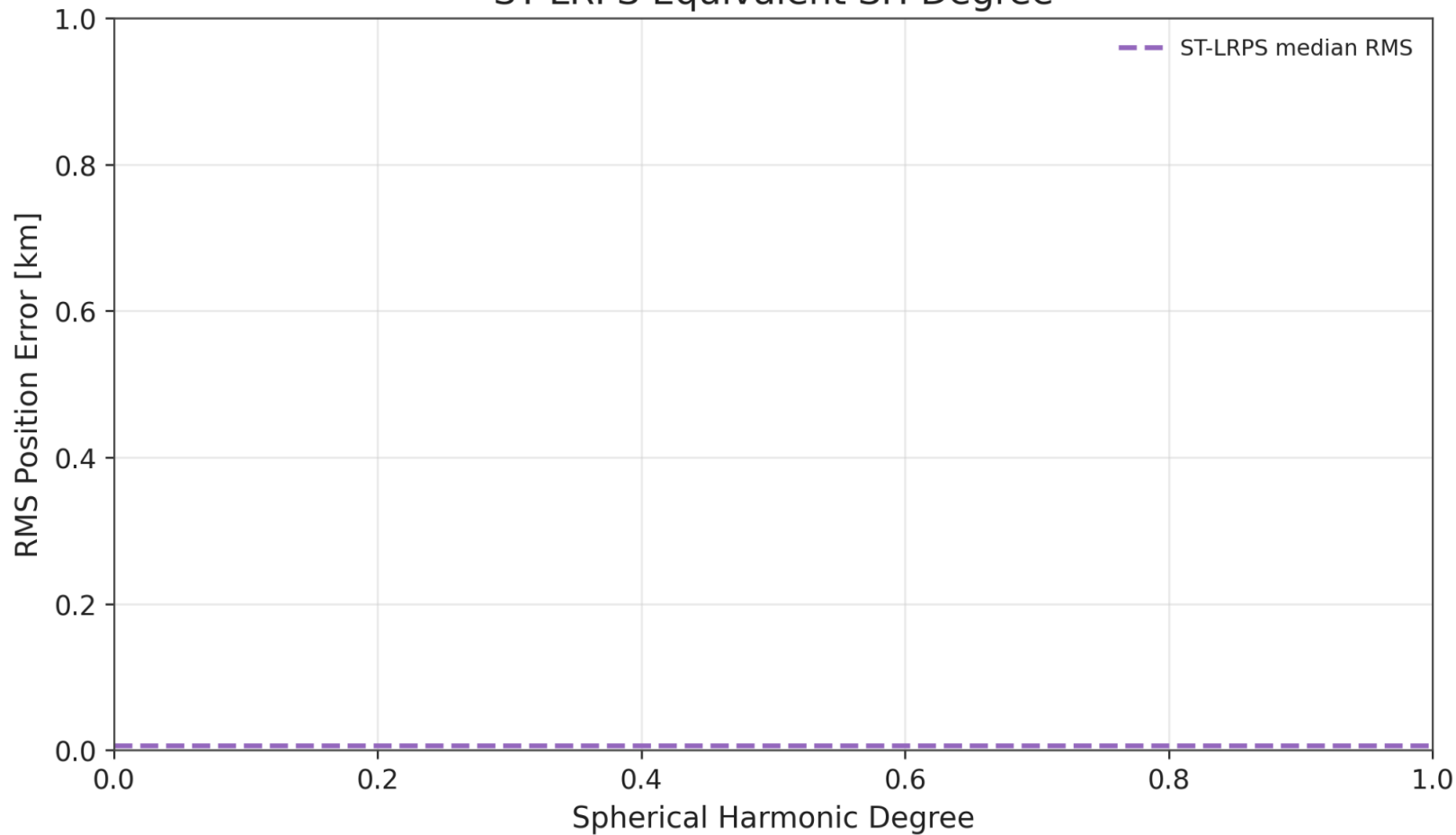
Runtime vs Accuracy



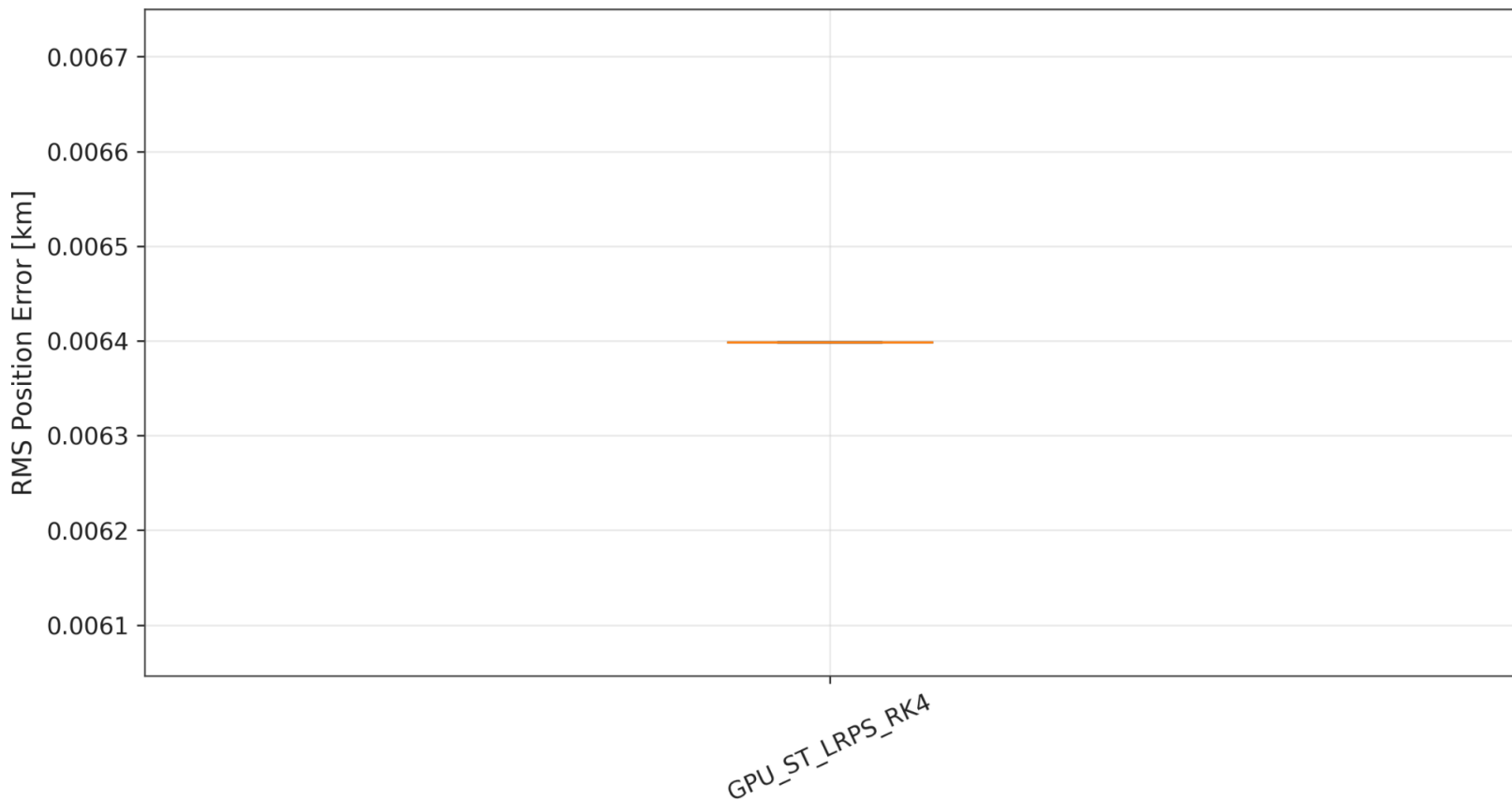
GPU RK4 Accuracy Ranking vs SH200 DOP853



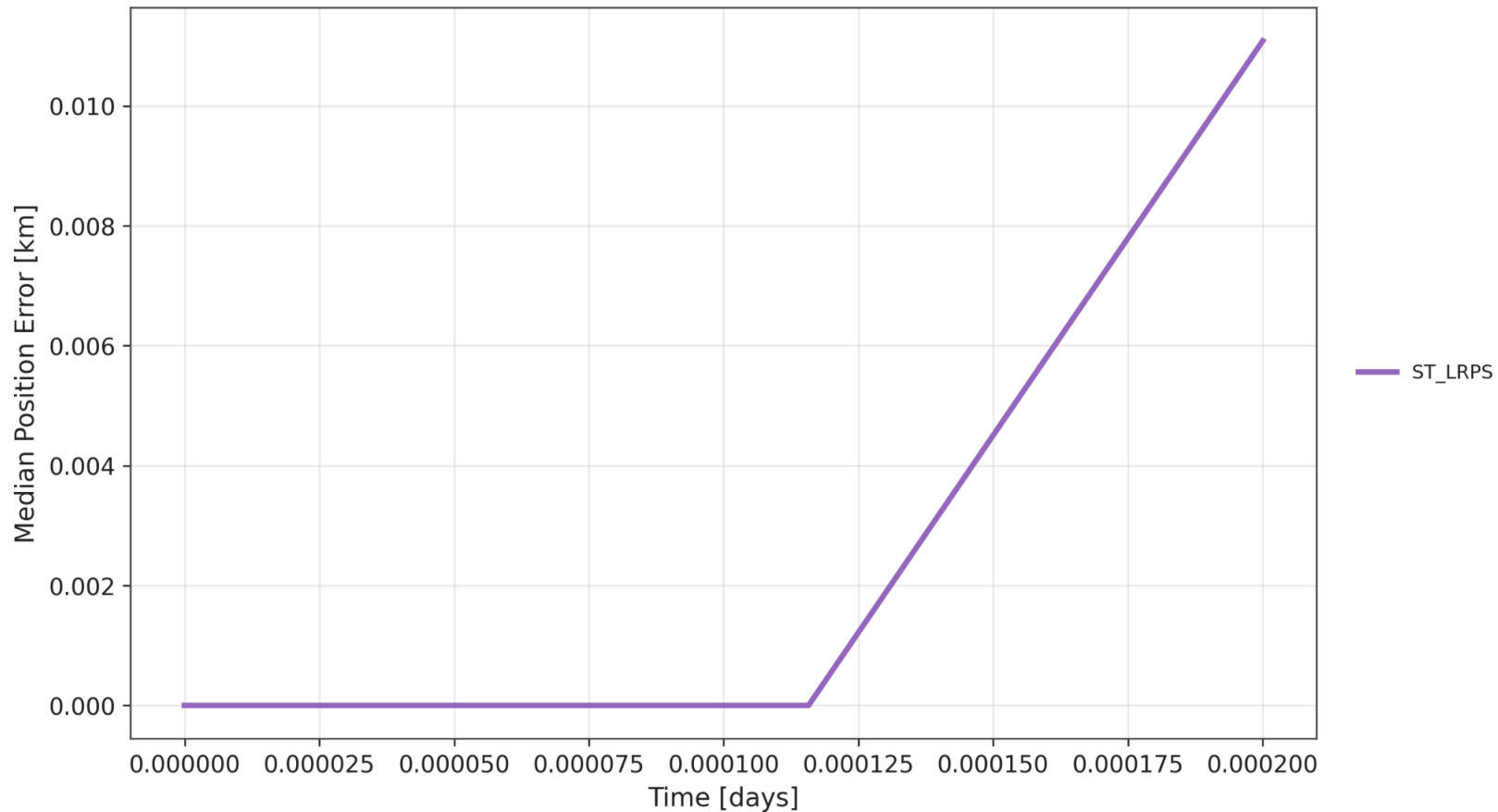
ST-LRPS Equivalent SH Degree



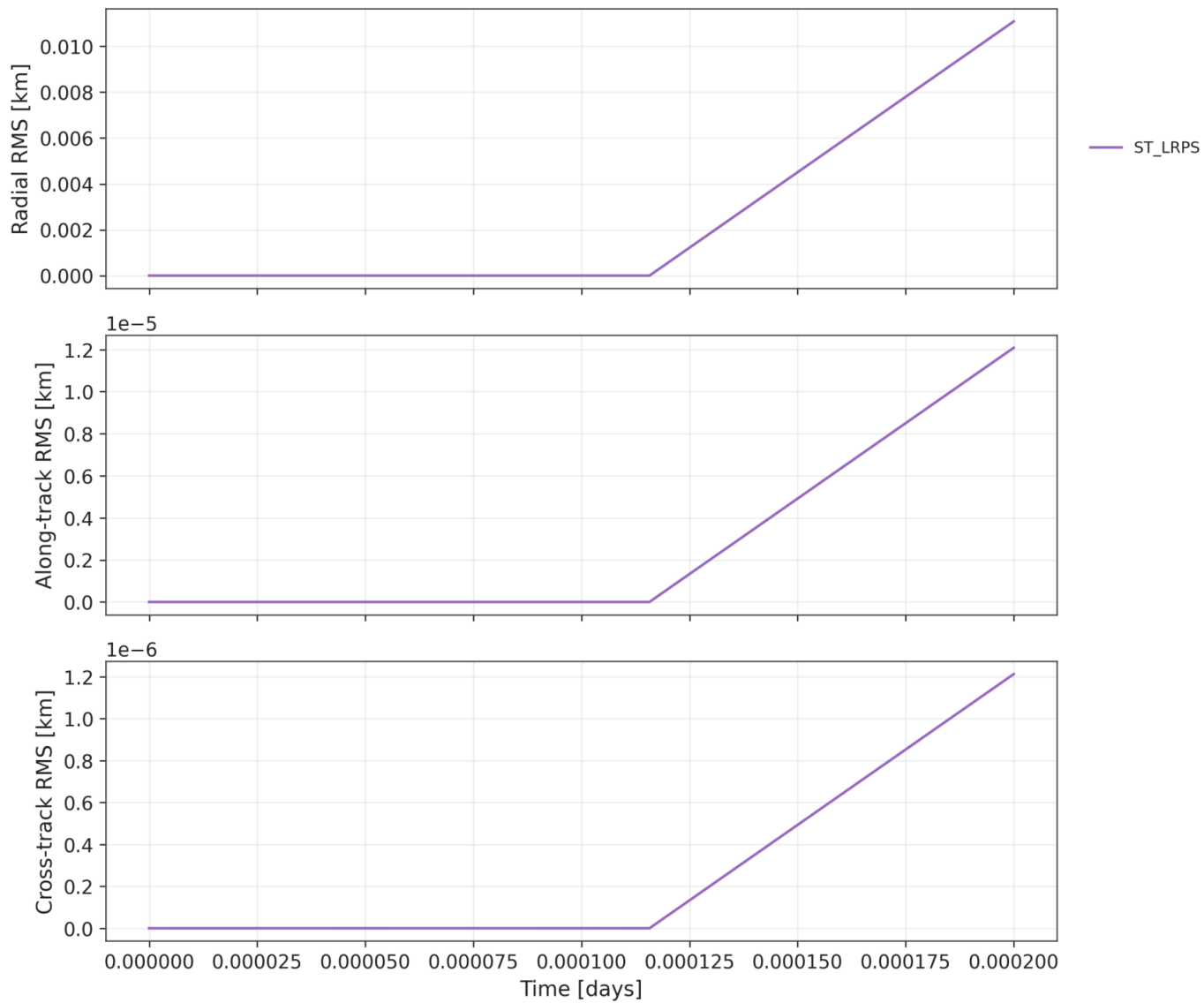
RMS Error Distribution Across Scenarios



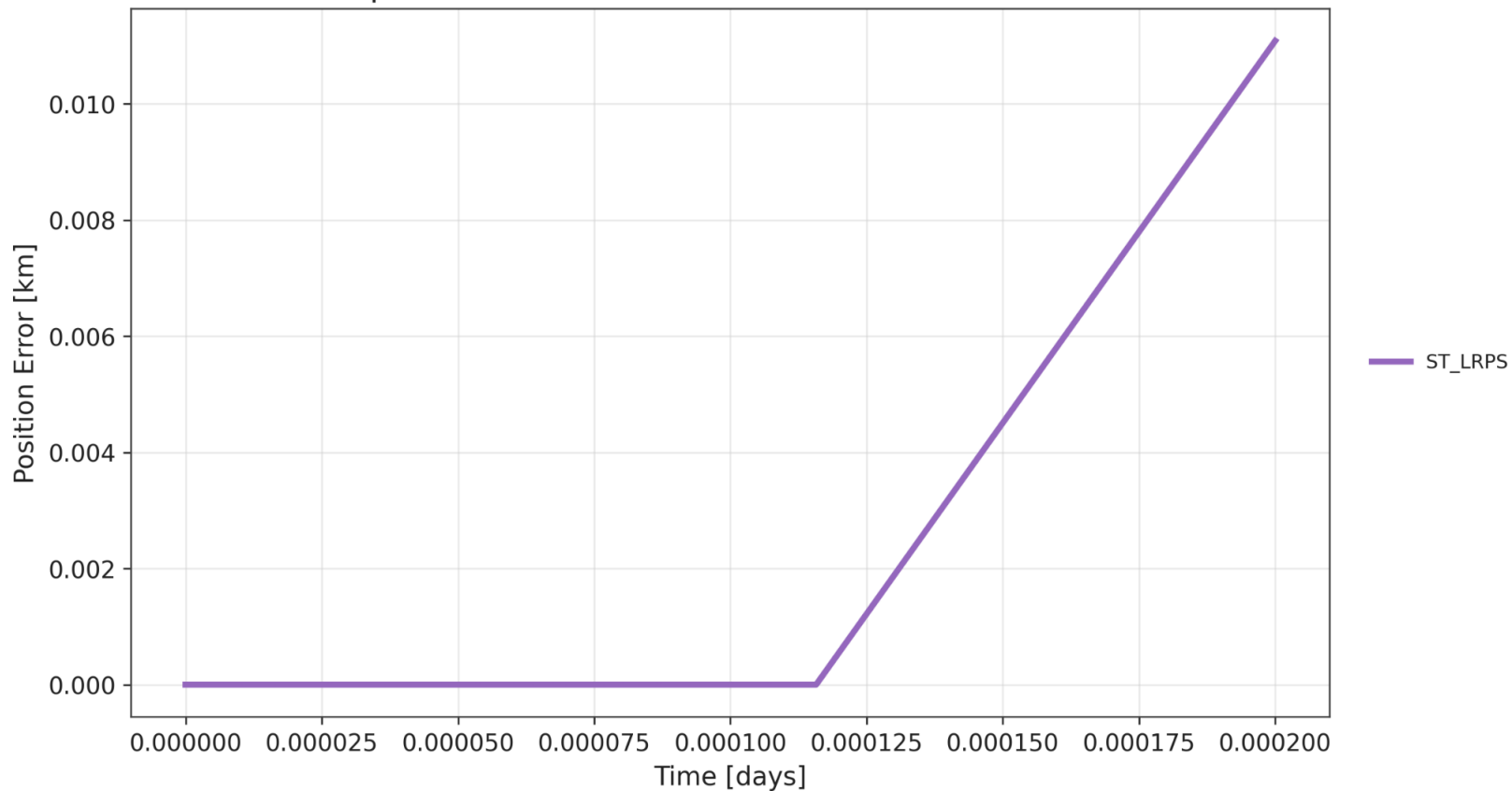
Ensemble Position Error vs Time



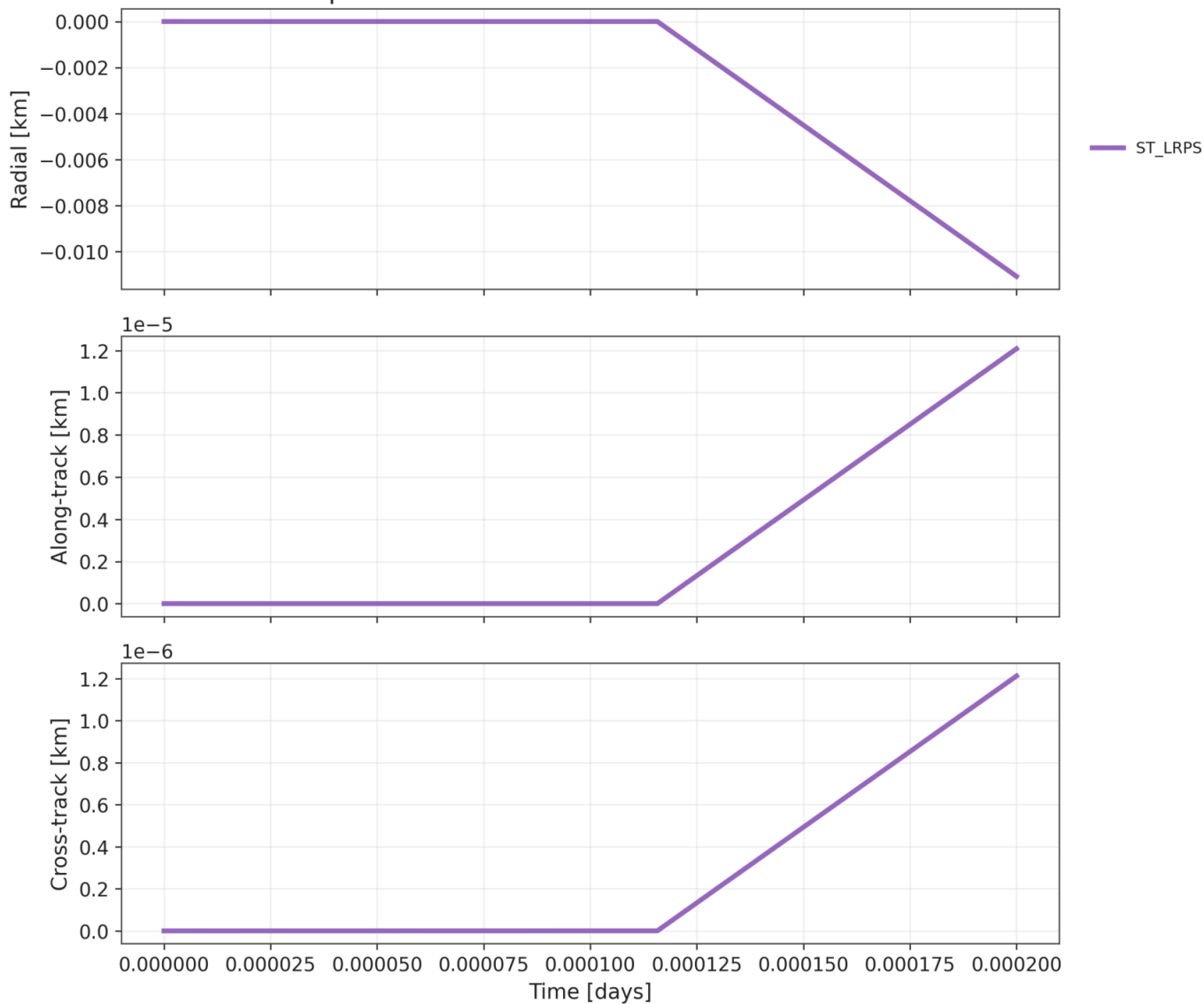
Ensemble RIC RMS Error vs Time



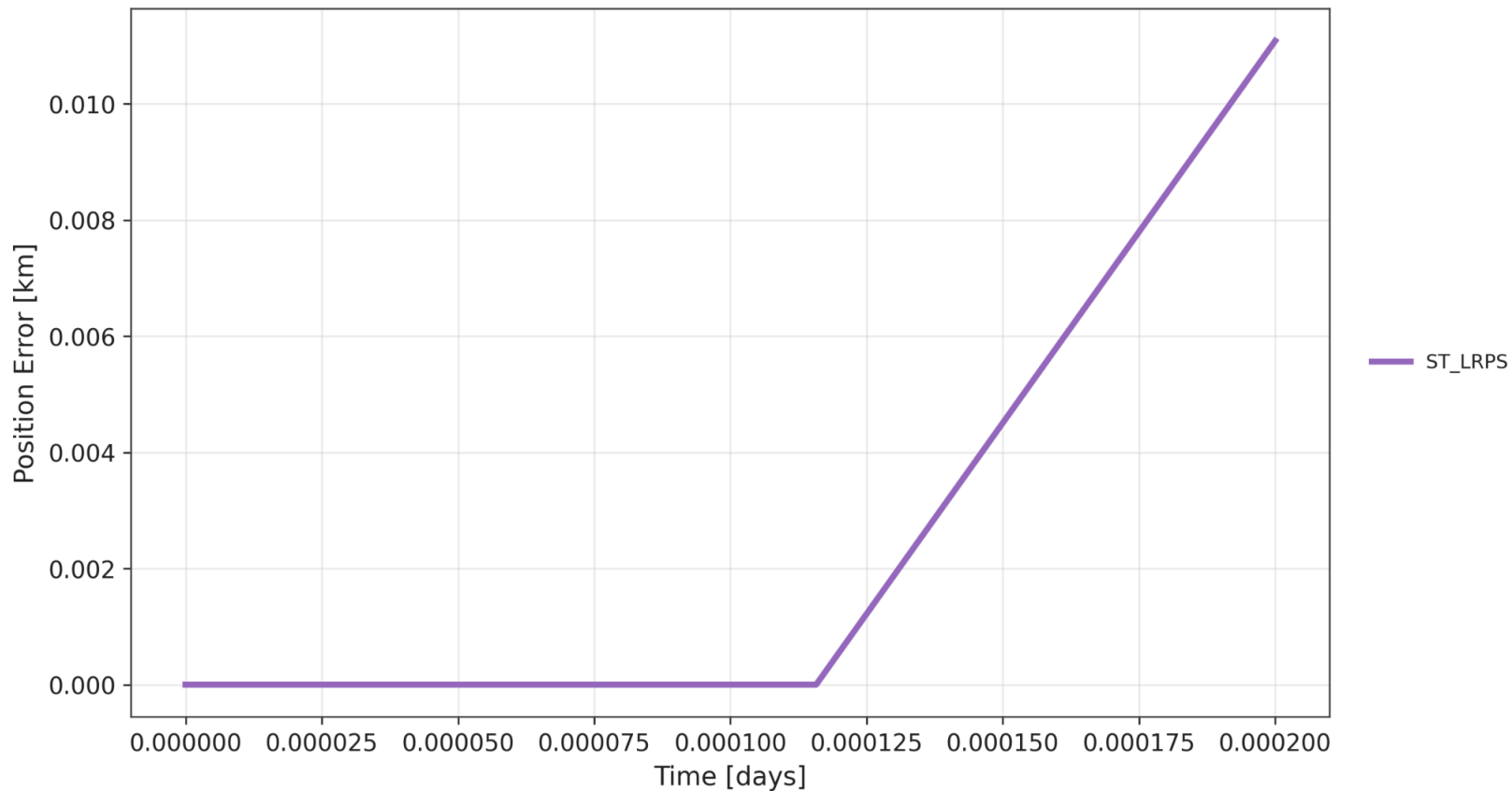
Representative ST-LRPS Scenario 0: Position Error



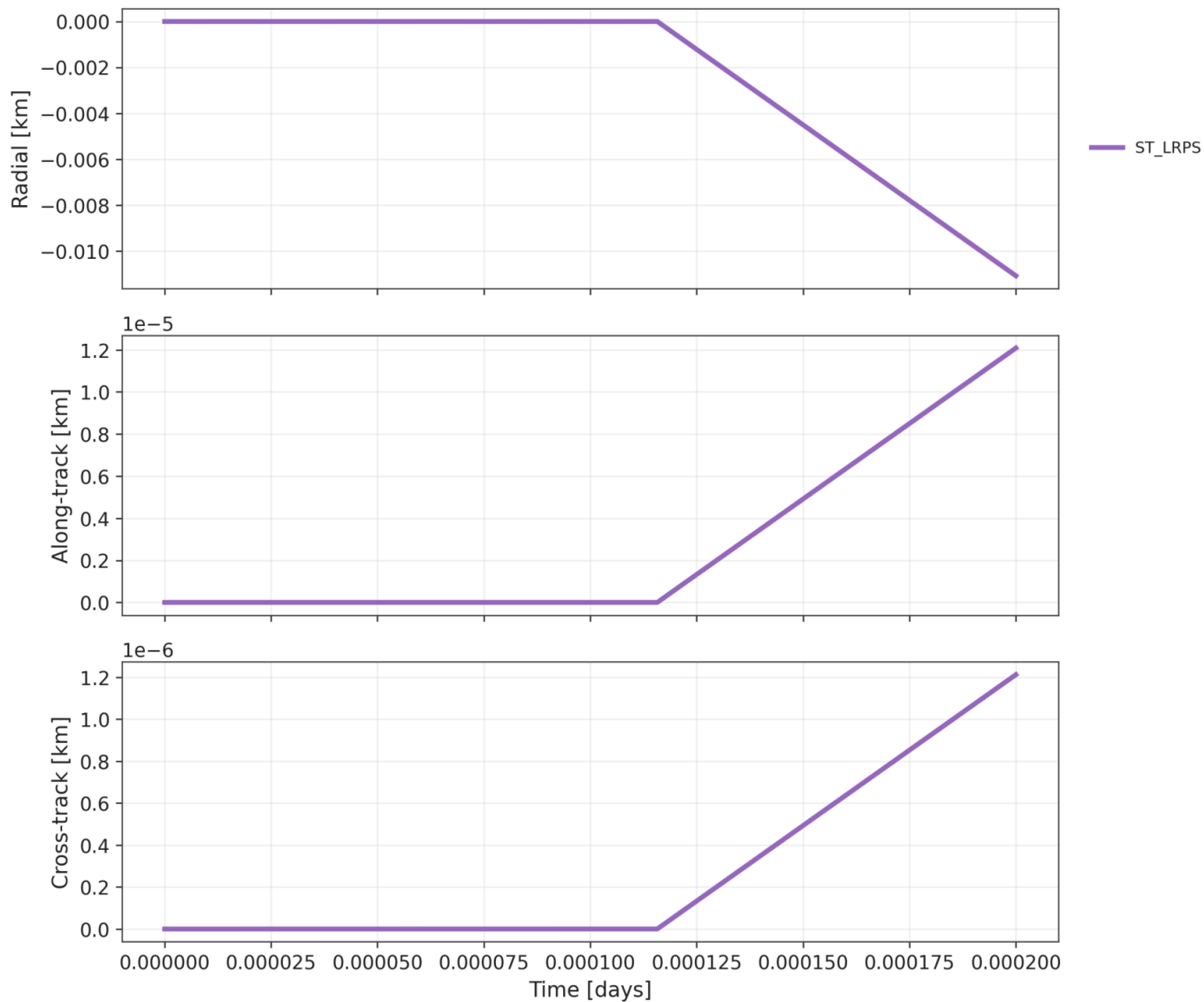
Representative ST-LRPS Scenario 0: RIC Error



Worst ST-LRPS Scenario 0: Position Error



Worst ST-LRPS Scenario 0: RIC Error



Notes

Notes / warnings:

- SH200 DOP853 is a high-accuracy numerical reference, not physical ground truth.
- GPU SH200 RK4 vs SH200 DOP853 estimates fixed-step RK4 and framework error.
- Lower-degree GPU SH models include both truncation and RK4/framework error.
- ST-LRPS includes surrogate-model error plus RK4/framework error.
- Frame mode: `match_dynamics_engine`